Summary in Graph

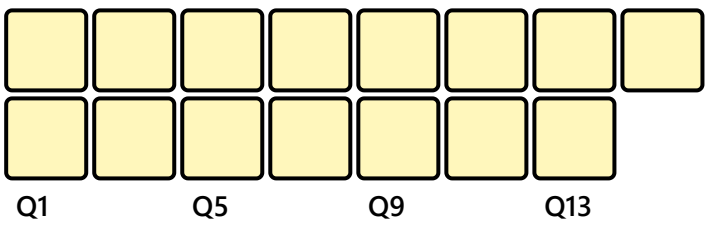
Exam Summary (GO Classes GATE CS/DA | Linear Algebra | Weekly Quiz 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (71) out of 714 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Linear Algebra

Let A be an $n \times n$ matrix of real or complex numbers. Which of the following statements are equivalent to: "the matrix A is invertible"?

A. The columns of A are linearly independent.
B. The rows of A are linearly independent.
C. The only solution of the homogeneous equations $Ax = 0$ is $x = 0$.

D. The rank of A is n.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let M be a 2×2 matrix with the property that the sum of the elements of each row and sum of the elements of each column is the same constant c. Which (if any) any of the vectors must be an eigenvector of M ?

$$U = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, V = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, W = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

A. U
B. V
C. W
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A and B be two $n \times n$ matrices. If B is invertible and $(I + BA)^{-1} = 2B^2$, then which of the following is the correct definition of A in terms of B?

A. $A = (1/2)B^{-3} - B^{-1}$
B. $A = 2B^{-3} - B^{-1}$
C. $A = 2B^3 - I$
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

A 4×4 matrix A has rank 3 . Which of the following is/are true?

1. A^{-1} does not exist
2. A^{-1} may exist, and if it does, its rank must be less than 3
3. A^{-1} may exist, and if it does, its rank must be greater than 1
4. A^{-1} may exist, and it can take any rank less than 5

A. Only 1 is correct
B. Only 2,3 are correct
C. Only 4 is correct
D. None of the statements are correct.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider two matrices $\mathrm{A}_{6\times 3}$ and $\mathrm{B}_{3\times 6}$, the non zero eigenvalues(EVs) of matrix AB are $3, 2, 7, 8$; see the following statements

$\mathrm{S1}$: The EVs of BA must be all 0 for the above case to be true

$\mathrm{S2}$: AB should not have more than 3 non zero EVs

A. $\mathrm{S1}$ is true, $\mathrm{S2}$ is false
B. $\mathrm{S2}$ is true, $\mathrm{S1}$ is false
C. Both $\mathrm{S1}$ and $\mathrm{S2}$ are true
D. Neither $\mathrm{S1}$ nor $\mathrm{S2}$ is true

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6

Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let the $n \times n$ matrix A have an eigenvalue λ with corresponding eigenvector v.
Which of the following statements are true for matrix A.

A. $-v$ is an eigenvector of $-A$ with eigenvalue $-\lambda$.
B. If v is also an eigenvector of the $n \times n$ matrix B with eigenvalue μ, then $\lambda\mu$ is an eigenvalue of AB.
C. Let μ be an eigenvalue of the $n \times n$ matrix B corresponding to the eigenvector v, Then $\lambda + \mu$ is an eigenvalue of $A + B$.
D. eigenvalue of A^3 is λ^3 and the eigenvector is v^3 .

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7

Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Consider the following matrix A:

$$\begin{bmatrix} 2 & -1 & 0 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Which of the following regarding the matrix A is/are correct?

A. $\lambda = 2$ is one of the eigenvalues of A.
B. $\lambda = 1$ is one of the eigenvalues of A.
C. If x is the eigenvector corresponding to eigenvalue λ then A^3x is equivalent to λ^3x i.e. $A^3x = \lambda^3x$.
D. If x is the eigenvector corresponding to eigenvalue λ of A then x is also the eigenvector of A^{-1}.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Suppose a 3×5 matrix A has rank $r = 3$. Then the equation $Ax = b$ $\mathbf{BLANK\ 1}$ has $\mathbf{BLANK\ 2}$.
Which of the following are appropriate words that can be filled in $\mathbf{BLANK\ 1}$ has $\mathbf{BLANK\ 2}$.?

A. BLANK 1: Always, BLANK 2: Unique solution
B. BLANK 1: Always, BLANK 2: Infinitely many solutions
C. BLANK 1: Sometimes, BLANK 2: Unique solution
D. BLANK 1: Sometimes, BLANK 2: Infinitely many solutions

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 04sec | Discuss

Q #9 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following statements is/are $\mathbf{NOT\ CORRECT}$?

A. If $v1$ and $v2$ are linearly independent eigenvectors then they can correspond to the same eigenvalue.
B. If A is a nilpotent matrix, meaning that $A^k = 0$ for the least positive integer k, and if $k = 2$, then A can have non-zero value as its eigenvalue.
C. λ is the eigenvalue of A if and only if λ is the eigenvalue of A^T.
D. If λ is an eigenvalue of an invertible matrix A then λ^{-1} is an eigenvalue of A^{-1}.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Given an $m \times n$ matrix A whose rows are linearly independent. Now, consider following statements regarding A:

$S1$: The system of equations $Ax = b$ for any b is consistent.
$S2$: $Ax = b$ always has a unique solution.

Which of the following is $\mathbf{TRUE}$ regarding $S1$ and $S2$?

A. Both $S1$ and $S2$ are TRUE.
B. $S1$ is TRUE and $S2$ is FALSE.
C. $S1$ is FALSE and $S2$ is TRUE.
D. Both $S1$ and $S2$ are FALSE.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following statements is/are $\mathbf{FALSE}$?

A. For $n \times n$ real-symmetric matrices A and B, AB and BA always have the same eigenvalues.
B. For $n \times n$ matrices A and B with B invertible, AB and BA always have the same eigenvalues.
C. For $n \times n$ matrices A and B with B invertible, AB and BA always have the same eigenvectors.
D. For $n \times n$ real-symmetric matrices A and B, AB and BA always have the same eigenvectors.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

The rank and nullity of a matrix A are 4 and 2 , respectively. The nullity of $A^\top$ is 3 . What are the dimensions of A ?

A. 6×7
B. 4×5
C. 7×6
D. 5×4

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the followings(s) is/are TRUE ?

A. If a system of linear equations has no free variables, then it has a unique solution.
B. If an augmented matrix $[A \mid b]$ is transformed into $[C \mid d]$ by elementary row operations, then the equations $Ax = b$ and $Cx = d$ have exactly the same solution sets.
C. If A is an $m \times n$ matrix and the equation $Ax = b$ is consistent for every b in $\mathbb{R}^m$, then A has m pivot columns.
D. If a matrix A with dimensions m by n has a pivot position in every row, then the equation $Ax = b$ has a unique solution for every b in the set of real numbers of dimension m.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following(s) is/are TRUE ?

A. If none of the vectors in the set $S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ in $\mathbb{R}^3$ is a multiple of one of the other vectors, then S is linearly independent.
B. If $\vec{u}, \vec{v}$, and $\vec{w}$ are nonzero vectors in $\mathbb{R}^2$, then $\vec{u}$ is a linear combination of $\vec{v}$ and $\vec{w}$.
C. If $\vec{w}$ is a linear combination of $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^n$, then $\vec{u}$ is a linear combination of $\vec{v}$ and $\vec{w}$.
D. Suppose that $\vec{v}_1, \vec{v}_2$, and $\vec{v}_3$ are non-zero vectors in $\mathbb{R}^5$, $\vec{v}_2$ is not a multiple of $\vec{v}_1$, and $\vec{v}_3$ is not a linear combination of $\vec{v}_1$ and $\vec{v}_2$. Then $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is linearly independent.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the following two statements:
$S1$. If $AB = I$, then A is invertible.

$S2$. If A is a 3×3 matrix and the equation $A\vec{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has a unique solution, then A is invertible.

Which of the following options is correct?

A. $S1$ is true but $S2$ is false
B. $\mathrm{S1}$ is false but $\mathrm{S2}$ is true
C. Both are true
D. Both are false

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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